

NONNULL DISTRIBUTION OF THE DETERMINANT OF B-STATISTIC IN MULTIVARIATE ANALYSIS

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Key words: Exact and asymptotic distributions; incomplete gamma
functions; matrix random variables; power computations.

Summary: In this paper exact distribution of the determinant of
B-statistic in multivariate analysis has been derived
in terms of incomplete gamma functions. Power compu-
tations of the test statistic are tabulated and compared
with those of de Waal and J. Roy. Asymptotic distri-
bution is also studied.

1. INTRODUCTION

Let the $p \times p$ random matrices A_1 and A_2 be independently distributed
according to $W_p(A_1; m, \Sigma)$ and $W_p(A_2; n, \Sigma, \theta)$ respectively, i.e. A_1 is
distributed as a central Wishart with m degrees of freedom, and A_2 is
distributed as a noncentral Wishart with n degrees of freedom and non-
centrality parameter θ of rank p . Let the $p \times p$ random matrix L be
defined as

$$L = (A_1 + A_2)^{-\frac{1}{2}} A_1 (A_1 + A_2)^{-\frac{1}{2}}.$$

STMA 1986 SUBJECT CLASSIFICATION: 03.080 (05.030)

The distribution of L is known in the literature as a noncentral multivariate beta type I (see de Waal (1968)).

The purpose of this paper is to study the nonnull distributions of $|L|$ and $|I - L|$. First, in section 2 some distributional properties of L are stated. The exact and asymptotic distributions of $|L|$ and $|I - L|$ have been derived in sections 3 and 4 respectively. Finally, in section 5 power computations of the test statistic are carried out and comparisons are made with the known results.

It may be noted that many authors have studied the distribution of random matrices, e.g. J. Roy (1965), Gupta (1971, 1975, 1977), Gupta and Chattopadhyay (1979), Gupta *et al.* (1975), and Gupta and Rathie (1983). The method used in this paper has been used previously by Nagarsenker (1979) and Tang (1981).

2. DISTRIBUTIONAL PROPERTIES OF L .

The hypergeometric function of a matrix argument S , used in this paper, is defined by

$$\begin{aligned} {}_pF_q(a_1, a_2, \dots, a_p; b_1, b_2, \dots, b_q; S) \\ = \sum_{k=0}^{\infty} \sum_{\kappa} \frac{(a_1)_{\kappa} (a_2)_{\kappa} \dots (a_p)_{\kappa} C_{\kappa}(S)}{(b_1)_{\kappa} (b_2)_{\kappa} \dots (b_q)_{\kappa} k!} \end{aligned}$$

where $C_{\kappa}(S)$, called zonal polynomials (see James (1964), Constantine (1963)) are homogeneous polynomials of order k in the latent roots of S ; the $p \times p$ matrix argument S is either symmetric or is a product of two symmetric matrices; $\kappa = \{k_1, k_2, \dots, k_p\}$, $k_1 \geq k_2 \geq \dots \geq k_p \geq 0$,

$\sum_{i=1}^p k_i = k$; $a_1, a_2, \dots, a_p$; $b_1, b_2, \dots, b_q$ are real or complex constants

with suitable restrictions on the b_j 's; $(c)_{\kappa} = \prod_{i=1}^p (c - (i-1)/2)_{k_i}$;

$$(c)_{\kappa} = \Gamma(c + k) / \Gamma(c).$$

De Waal (1968) derived the following two results which are stated without proof.

Lemma 2.1 If A_1 and A_2 are independently distributed as $W_p(A_1: m, \Sigma)$ and $W_p(A_2: n, \Sigma, \theta)$ respectively then the pdf of L is given by

$$(2.1) \quad f_L(L) = [\Gamma_p(m/2) \Gamma_p(n/2)]^{-1} |2\Sigma|^{-\frac{1}{2}(m+n)} \cdot \text{etr}(-\theta/2) \cdot |L|^{\frac{1}{2}(m-p-1)} \\ \cdot |I-L|^{\frac{1}{2}(n-p-1)} \cdot \int_{T>0} \text{etr}(-\frac{1}{2}\Sigma^{-1}T) |T|^{\frac{1}{2}(m+n-p-1)} \\ \cdot {}_0F_1(\frac{n}{2}; \frac{1}{4}\theta\Sigma^{-1}T^{\frac{1}{2}}(I-L)T^{\frac{1}{2}})dT, \quad 0 < L < I.$$

Lemma 2.2 If L is distributed according to a noncentral beta distribution of full rank, then

$$(2.2) \quad E|L|^h = \frac{\Gamma_p(\frac{m+n}{2}) \Gamma_p(\frac{m}{2}+h)}{\Gamma_p(\frac{m}{2}) \Gamma_p(\frac{m+n}{2}+h)} \text{etr}(-\theta/2) \cdot {}_1F_1(\frac{m+n}{2}; \frac{m+n}{2}+h; \frac{1}{2}\theta)$$

and

$$(2.3) \quad E|I-L|^h = \frac{\Gamma_p(\frac{m+n}{2}) \Gamma_p(\frac{n}{2}+h)}{\Gamma_p(\frac{n}{2}) \Gamma_p(\frac{m+n}{2}+h)} \text{etr}(-\theta/2) \\ \cdot {}_2F_2(\frac{m+n}{2}, \frac{n}{2}+h; \frac{n}{2}, \frac{m+n}{2}+h; \frac{1}{2}\theta).$$

The following expansion due to Barnes (1899) of log gamma function will also be needed in the sequel.

Lemma 2.3 For bounded h ,

$$(2.4) \quad \ln \Gamma(x+h) = \ln \sqrt{2\pi} + (x+h-\frac{1}{2}) \ln x - x \\ + \sum_{r=1}^m \frac{x^{-r}(-1)^{r+1}}{r(r+1)} B_{r+1}(h) + R_{m+1}(h)$$

where $R_m(x)$ is the remainder such that $|R_m(x)| \leq a/|x|^m$, a is a constant independent of x , and $B_r(h)$ is the Bernoulli polynomial of

order r defined by $e^{h\tau}(e^\tau - 1)^{-1} = \sum_{r=0}^{\infty} \frac{\tau^r}{r!} B_r(h)$.

3. EXACT DISTRIBUTIONS OF $|L|$ AND $|I - L|$.

Let $L^* = -(m+n-u)q \ln |L|$, $0 < L^* < \infty$, where u and q (> 0) are constants to be specified later (see Nagarsenker (1979), Gupta *et al.* (1975)). Then the exact distribution of L^* is given by the following theorem.

Theorem 3.1 If L is distributed according to a noncentral beta distribution of full rank, then the pdf of L^* is given by

$$(3.1) \quad f_{L^*}(x) = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p(\frac{m+n}{2}, \kappa)}{\Gamma_p(\frac{m}{2})} \frac{C_{\kappa}(\theta/2)}{k!} \\ \cdot \sum_{r=0}^{\infty} \frac{A_r^{\kappa}}{(\frac{m+n-u}{2})^{r+\nu}} g_{r+\nu}(2q, x), \quad 0 < x < \infty$$

and the cdf of L^* is given by

$$(3.2) \quad P[L^* \leq x] = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p(\frac{m+n}{2}, \kappa)}{\Gamma_p(\frac{m}{2})} \frac{C_{\kappa}(\theta/2)}{k!} \\ \cdot \left[\sum_{r=0}^{\infty} \frac{A_r^{\kappa}}{(\frac{m+n-u}{2})^{r+\nu}} G_{r+\nu}(2q, x) \right]$$

where $\kappa = \{k_1 \geq k_2 \geq \dots \geq k_p\}$, $\sum_i k_i = k$, denotes a partition of k ;

$\sum_{\kappa}$ denotes the summation over all partitions κ of k ; $C_{\kappa}(\cdot)$ is the corresponding zonal polynomial; $\nu = np/2 + k$; $g_{\alpha}(2q, x)$ is the gamma

density $[(2q)^{\alpha} \Gamma(\alpha)]^{-1} x^{\alpha-1} \exp(-x/2q)$; $G_{\alpha}(2q, x) = \int_0^x g_{\alpha}(2q, t) dt$; and A_r^{κ}

is defined recursively by

$$(3.3) \quad A_r^K = \frac{1}{r} \sum_{\ell=1}^r \ell Q_\ell A_{r-\ell}^K, \quad A_0^K = 1, \quad r = 1, 2, \dots;$$

$$Q_\ell = \frac{(-1)^{\ell-1}}{\ell(\ell+1)} \sum_{j=1}^p \left[B_{\ell+1} \left(\frac{1-j-n+u}{2} \right) - B_{\ell+1} \left(k_j + \frac{1-j+u}{2} \right) \right], \quad \ell = 1, 2, \dots.$$

Proof Let $\phi_L^*(t)$ denote the characteristic function of L^* . Then

$$\begin{aligned} \phi_L^*(t) &= E[\exp(iL^*t)] \\ &= E[|L|^{-(m+n-u)qit}] \end{aligned}$$

where $E(\cdot)$ is the expectation operator, and $i = \sqrt{-1}$. Using lemma 2.2, we get

$$\phi_L^*(t) = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p(\frac{m+n}{2}, \kappa)}{\Gamma_p(\frac{m}{2})} \frac{C(t, \kappa)}{k!} C_{\kappa}(\theta/2)$$

where

$$\begin{aligned} (3.4) \quad C(t, \kappa) &= \frac{\Gamma_p(\frac{m}{2} - (m+n-u)qit)}{\Gamma_p(\frac{m+n}{2} - (m+n-u)qit, \kappa)} \\ &= \prod_{j=1}^p \left\{ \frac{\Gamma[(\frac{m+n-u}{2})(1-2qit) + \frac{1-j-n+u}{2}]}{\Gamma[(\frac{m+n-u}{2})(1-2qit) + \frac{1-j+u}{2} + k_j]} \right\}. \end{aligned}$$

Apply Barnes' formula to (3.4) and get

$$\begin{aligned} \ln C(t, \kappa) &= -\left(\frac{np}{2} + k\right) \ln \left[\frac{1}{2}(m+n-u)(1-2qit) \right] \\ &\quad + \sum_{r=1}^{\infty} Q_r \left[\frac{1}{2}(m+n-u)(1-2qit) \right]^{-r} \end{aligned}$$

where Q_r is given in (3.3). Hence

$$C(t, \kappa) = \left[\frac{1}{2}(m+n-u)(1-2qit) \right]^{-\left(\frac{np}{2} + k\right)} \\ \cdot \left[1 + \sum_{r=1}^{\infty} A_r^{\kappa} \left[\frac{1}{2}(m+n-u)(1-2qit) \right]^{-r} \right]$$

where A_r^{κ} is given in (3.3). Thus, letting $\nu = \frac{np}{2} + k$, we get,

$$(3.5) \quad \phi_{L^*}(t) = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p\left(\frac{m+n}{2}, \kappa\right)}{\Gamma_p\left(\frac{n}{2}\right)} \frac{C_{\kappa}(\theta/2)}{k!} \left[\left\{ \frac{m+n-u}{2} \right\}^{-\nu} (1-2qit)^{-\nu} \right. \\ \left. + \sum_{r=1}^{\infty} A_r^{\kappa} \left[\frac{1}{2}(m+n-u)(1-2qit) \right]^{-(r+\nu)} \right].$$

Now by inverting the characteristic function (3.5), recognizing that $(1-2qit)^{-\alpha}$ is the characteristic function of the gamma density $g_{\alpha}(2q, x)$, we obtain the exact pdf of L^* and its exact cdf as given in (3.1) and (3.2) respectively,

Next the exact distribution of $L^{**} = -(m+n-u)qLn|I-L|$, where u and $q(>0)$ are constants, is given by the following theorem.

Theorem 3.2 If L is distributed as a noncentral beta of full rank, then the pdf and cdf of L^{**} are respectively given by

$$(3.6) \quad f_{L^{**}}(x) = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p\left(\frac{m+n}{2}, \kappa\right)}{\Gamma_p\left(\frac{n}{2}, \kappa\right)} \frac{C_{\kappa}(\theta/2)}{k!} \\ \cdot \sum_{r=0}^{\infty} D_r \left(\frac{m+n-u}{2} \right)^{-(r+\eta)} g_{r+\eta}(2q, x), \quad 0 < x < \infty,$$

and

$$(3.7) \quad P[L^{**} \leq x] = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p\left(\frac{m+n}{2}, \kappa\right)}{\Gamma_p\left(\frac{n}{2}, \kappa\right)} \frac{C_{\kappa}(\theta/2)}{k!} \\ \cdot \sum_{r=0}^{\infty} D_r \left[\frac{m+n-u}{2} \right]^{-(r+\eta)} G_{r+\eta}(2q, x)$$

where $\eta = \frac{mp}{2}$,

$$D_r = \frac{1}{r} \sum_{\ell=1}^r \ell R_{\ell} D_{r-\ell}, \quad D_0 = 1, \quad r = 1, 2, \dots;$$

$$(3.8) \quad R_{\ell} = \frac{(-1)^{\ell-1}}{\ell(\ell+1)} \sum_{j=1}^p \left[B_{\ell+1} \left(k_j + \frac{1-j-m+u}{2} \right) - B_{\ell+1} \left(k_j + \frac{1-j+u}{2} \right) \right]$$

$$\ell = 1, 2, \dots$$

Proof As in the proof of theorem 3.1, from lemma 2.2 we obtain the characteristic function of L^{**} as

$$\phi_{L^{**}}(t) = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \left[\frac{\Gamma_p(\frac{m+n}{2}, \kappa)}{\Gamma_p(\frac{n}{p}, \kappa)} \frac{C_{\kappa}(\theta/2)}{k!} \right] E(t, \kappa),$$

where

$$E(t, \kappa) = \frac{\Gamma_p(\frac{n}{2} - \frac{1}{2}(m+n-u)2qit, \kappa)}{\Gamma_p(\frac{m+n}{2} - \frac{1}{2}(m+n-u)2qit, \kappa)}.$$

From lemma 2.3, after some algebra (see Javier (1982) for details) we get

$$(3.9) \quad \phi_{L^{**}}(t) = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{\Gamma_p(\frac{m+n}{2}, \kappa)}{\Gamma_p(\frac{n}{2}, \kappa)} \frac{C_{\kappa}(\theta/2)}{k!}$$

$$\cdot \sum_{r=0}^{\infty} D_r \left[\frac{1}{2}(m+n-u)(1-2qit) \right]^{-r} \eta^r.$$

Inverting (3.9) gives the result.

4. ASYMPTOTIC DISTRIBUTIONS OF $|L|$ AND $|I-L|$

In this section we derive an asymptotic expansion of the distribution of a suitable function of each of $|L|$ and $|I-L|$.

Theorem 4.1 The asymptotic distribution of $L^* = -(m+n-u)q\eta|L|$ is given by

$$\begin{aligned}
 (4.1) \quad P[L^* \leq x] &= \sum_{k=0}^{\infty} P(k) G_{\nu}(2q, x) \\
 &+ \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} w_{1\kappa} [G_{1+\nu}(2q, x) - G_{\nu}(2q, x)] \\
 &\cdot \frac{C_{\kappa}(\theta/2)}{k!} + O(m^{-2})
 \end{aligned}$$

where $\nu = \frac{np}{2} + k$, $P(k) = \text{etr}(-\theta/2) \frac{[\text{tr}(\theta/2)]^k}{k!}$.

$$w_{1\kappa} = -\frac{1}{\rho} \left[k(1-\rho + \frac{n}{m}) + \frac{1}{m} \left(\sum_{j=1}^p k_j^2 - \sum_{j=1}^p j k_j \right) \right]; \text{ and } \rho = 1 + \frac{n-p-1}{2m}.$$

Proof In the expression for the cdf of L^* let

$$(4.2) \quad \psi(\kappa) = \frac{\Gamma_p(\frac{m+n}{2}, \kappa)}{\Gamma_p(\frac{m}{2})} = \prod_{j=1}^p \frac{\Gamma(\frac{m+n-u}{2} + k_j + \frac{1-j+u}{2})}{\Gamma(\frac{m+n-u}{2} + \frac{1-j+u-n}{2})}.$$

Using Barnes' formula from lemma 2.3, we have

$$\ln \psi(\kappa) = (\frac{np}{2} + k) \ln(\frac{m+n-u}{2}) + \sum_{r=1}^{\infty} Q'_r (\frac{m+n-u}{2})^r,$$

where for $r = 1, 2, \dots$

$$Q'_r = \frac{(-1)^{r+1}}{r(r+1)} \sum_{j=1}^p \left[B_{r+1}(k_j + \frac{1-j+u}{2}) - B_{r+1}(\frac{1-j+u-n}{2}) \right].$$

Thus

$$(4.3) \quad \psi(\kappa) = (\frac{m+n-u}{2})^{\frac{np}{2} + k} \left[1 + \sum_{r=1}^{\infty} A'_r (\frac{m+n-u}{2})^{-r} \right]$$

where A'_r is defined recursively by

$$A'_r = \frac{1}{r} \sum_{\ell=1}^r \ell Q'_\ell A'_{r-\ell}, \quad A'_0 = 1, \quad r = 1, 2, \dots$$

Substituting in (3.2) from (4.3), the cdf of L^* can be written as

$$(4.4) \quad P[L^* \leq x] = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \left[\sum_{r=0}^{\infty} A_r^{\kappa} \left(\frac{m+n-u}{2} \right)^{-r} \right] \\ \cdot \left[\sum_{r=0}^{\infty} A_r^{\kappa} \left(\frac{m+n-u}{2} \right)^{-r} G_{r+\nu}(2q, x) \right] \frac{C_{\kappa}(\theta/2)}{k!}.$$

Multiplying out the two infinite series in square brackets in the last equation we get the product

$$G_{\nu}(2q, x) + [A_1^{\kappa} G_{\nu}(2q, x) + A_1^{\kappa} G_{1+\nu}(2q, x)] \left(\frac{m+n-u}{2} \right)^{-1} + O(m^{-2}) \\ = G_{\nu}(2q, x) + A_1^{\kappa} [G_{1+\nu}(2q, x) - G_{\nu}(2q, x)] \left(\frac{m+n-u}{2} \right)^{-1} + O(m^{-2}),$$

since $A_1^{\kappa} = -A_1^{\kappa}$ from equations (3.3) and (4.3). Thus (4.4) becomes

$$(4.5) \quad P[L^* \leq x] = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{C_{\kappa}(\theta/2)}{k!} G_{\nu}(2q, x) \\ + \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{A_1^{\kappa}}{\left(\frac{m+n-u}{2} \right)} [G_{1+\nu}(2q, x) - G_{\nu}(2q, x)] \\ \frac{C_{\kappa}(\theta/2)}{k!} + O(m^{-2}) \\ = \sum_{k=0}^{\infty} P(k) G_{\nu}(2q, x) \\ + \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{A_1^{\kappa}}{\left(\frac{m+n-u}{2} \right)} [G_{1+\nu}(2q, x) - G_{\nu}(2q, x)] + O(m^{-2}).$$

What remains to be done is the evaluation of A_1^{κ} , where

$A_1^{\kappa} = \frac{1}{2} \sum_{j=1}^p \left[B_2 \left(\frac{1-j-n+u}{2} \right) - B_2 \left(k_j + \frac{1-j+u}{2} \right) \right]$. Since $B_2(z) = z^2 - z + \frac{1}{6}$ (Anderson (1984), p. 312), after some algebra, we obtain

$$(4.6) \quad A_1^{\kappa} = \frac{1}{2} \left[- \left(\sum_{j=1}^p k_j^2 - \sum_{j=1}^p j k_j \right) + \left(\frac{n^2 p}{4} + \frac{np(p+1)}{4} - \frac{nup}{2} \right) - ku \right].$$

Now, choose u so that $\frac{n^2 p}{4} + \frac{np(p+1)}{4} - \frac{nup}{2} = 0$, and choose a number ρ'

such that $(m+n-u)q = \rho'm$. Obtaining the solution $u = \frac{1}{2}(n+p+1)$ and $\rho' = (1 + \frac{n-p-1}{2})q$. Substituting these values in (4.6) we get

$$A_1^K = -\frac{m}{2} \left[\frac{1}{m} \left(\sum_{j=1}^p k_j^2 - \sum_{j=1}^p j k_j \right) + k \left(1 - \frac{\rho'}{q} + \frac{n}{m} \right) \right].$$

Substituting this last expression in (4.5) gives the desired formula for the cdf of L^* .

Remark 4.1 When $q=1$, theorem 4.1 above reduces to de Waal's theorem (1968, p. 80).

Theorem 4.2 The asymptotic distribution of $L^{**} = -(m+n-u)q \ln|I-L|$ is given by

$$(4.7) \quad P[L^{**} \leq x] = G_\eta(2q, x) + \left[\frac{p(p+1)}{4} - \frac{np}{2} - 2\text{tr}(\theta/2) \right] \\ \cdot [G_{1+\eta}(2q, x) - G_\eta(2q, x)] + O(m^{-2})$$

where $\eta = \frac{mp}{2}$, $0 < q < \infty$.

Proof In the expression (3.7) for the cdf of L^{**} let

$$\bar{\psi}_\kappa = \frac{\Gamma_p(\frac{m+n}{2}, \kappa)}{\Gamma_p(\frac{n}{2}, \kappa)} = \prod_{j=1}^p \frac{\Gamma(\frac{m+n-u}{2} + k_j + \frac{1-j+u}{2})}{\Gamma(\frac{m+n-u}{2} + k_j + \frac{1-j-m+u}{2})}.$$

By Barnes' formula, the asymptotic expansion of this quantity is

$$(4.8) \quad \bar{\psi}_\kappa = \left(\frac{m+n-u}{2} \right) \left[1 + \sum_{r=1}^{\infty} \frac{D_r^1}{\left(\frac{m+n-u}{2} \right)^r} \right],$$

where $D_r^1 = \frac{1}{r} \sum_{\ell=1}^r \mathcal{L} R^1 D_{r-\ell}^1$, $D_0^1 = 1$, and

$$R_r^1 = \sum_{j=1}^p \frac{(-1)^{r+1}}{r(r+1)} \left[B_{r+1}(k_j + \frac{1-j+u}{2}) - B_{r+1}(k_j + \frac{1-j-m+u}{2}) \right], \quad r = 1, 2, \dots$$

Substituting in (3.7) from (4.8) we get

$$P[L^{**} \leq x] = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{C_{\kappa}(\theta/2)}{k!} \cdot \left[\sum_{r=0}^{\infty} \frac{D_r'}{\left(\frac{m+n-u}{2}\right)^r} \right] \\ \cdot \left[\sum_{r=0}^{\infty} \frac{D_r}{\left(\frac{m+n-u}{2}\right)^r} G_{r+\eta}(2q, x) \right].$$

Multiplying out the two infinite series enclosed in square brackets, the product is

$$G_{\eta}(2q, x) + [D_1' G_{\eta}(2q, x) + D_1' G_{1+\eta}(2q, x)] \cdot \left(\frac{m+n-u}{2}\right)^{-1} + O(m^{-2}) \\ = G_{\eta}(2q, x) + D_1 [G_{1+\eta}(2q, x) - G_{\eta}(2q, x)] \left(\frac{m+n-u}{2}\right)^{-1} + O(m^{-2})$$

$$\text{since } D_1' = R_1' = \frac{1}{2} \sum_{j=1}^p \left[B_2\left(k_j + \frac{1-j+u}{2}\right) - B_2\left(k_j + \frac{1-j-m+u}{2}\right) \right] = -R_1 = -D_1.$$

Hence

$$P[L^{**} \leq x] = \text{etr}(-\theta/2) \sum_{k=0}^{\infty} \sum_{\kappa} \frac{C_{\kappa}(\theta/2)}{k!} \left\{ G_{\eta}(2q, x) \right. \\ \left. + D_1 \left(\frac{m+n-u}{2}\right)^{-1} [G_{1+\eta}(2q, x) - G_{\eta}(2q, x)] \right\} + O(m^{-2}).$$

$$\text{But, } D_1 = \frac{1}{2} \sum_{j=1}^p \left[B_2\left(k_j + \frac{1-j-m+u}{2}\right) - B_2\left(k_j + \frac{1-j+u}{2}\right) \right] \\ = \frac{m}{2} [p(p+1)/4 - 2k - up/2].$$

Now setting $u = n$, we get

$$P[L^{**} \leq x] = G_{\eta}(2q, x) + [p(p+1)/4 - np/2 - 2\text{tr}(\theta/2)] \\ \cdot [G_{1+\eta}(2q, x) - G_{\eta}(2q, x)] + O(m^{-2}).$$

Remark 4.2 When n is such that $\frac{p(p+1)}{4} - \frac{np}{2} - 2\text{tr}(\theta/2) = 0$, and $q = 1$, the above result reduces to de Waal's theorem (1968, p. 82).

5. POWER CALCULATIONS IN THE LINEAR CASE

Consider the case when rank of the noncentrality parameter matrix θ is 1. Let λ be the only nonzero latent root of θ . In the literature, this is referred to as the linear case and exact power calculations have been done for $p=2$ by J. Roy (1965). Using his asymptotic formula involving a χ^2 -mixture, de Waal (1968) verified the accuracy of his results against Roy's exact figures. The general theorem 4.1 will now be specialized to the linear case. We need the following lemma.

Lemma 5.1 Let λ be the only nonzero latent root of a $p \times p$ symmetric matrix A . Then the zonal polynomial $C_\kappa(A)$ corresponding to a partition κ of k is given by

$$C_\kappa(A) = \begin{cases} \lambda^k & \text{if } \kappa = \{k\} \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 5.1 The asymptotic distribution of $L^* = -(m + \frac{n-p-1}{2})q \ln |L|$, where $|L|$ is the determinant of the generalized B-statistic L , in the linear case is given by

$$(5.1) \quad P[L^* \leq x] = \sum_{k=0}^{\infty} P(k) G_\nu(2q, x) + \exp(-\lambda/2) \sum_{k=0}^{\infty} w_{1k} \frac{(\lambda/2)^k}{k!} \\ \cdot [G_{1+\nu}(2q, x) - G_\nu(2q, x)] + O(m^{-2}),$$

where $P(k)$, $\tilde{G}_\alpha(\cdot, \cdot)$, ρ , and ν have been defined in theorem 4.1, and $w_{1k} = -\frac{1}{\rho} [k(1-\rho + \frac{n}{m}) + \frac{1}{m}(k^2 - k)]$ for $k = 1, 2, \dots$.

Proof Apply lemma 5.1 to the right-hand side of equation (4.1).

To check the accuracy of equation (5.1) a WATFIV program was written based on this equation to compute the power of $|L|$, linear case at 5% signifi-

cance level and with $p = 2$. The figures obtained are then compared with those obtained by de Waal (1968) and J. Roy (1965) and are presented in the tables below. Using the WATFIV program, a computer search was performed to find the optimum value of q . This value was found to be $q = 1,0025970$ and this value of q was used in the computation for the power. The threshold value was found according to the formula

$$\text{THRESV} = [1 + \frac{n}{m-1} \cdot EF]^{-2}$$

where EF is the right tail 5% point of an F -distribution with $D_1 = 2n$ and $D_2 = 2(m-1)$ degrees of freedom. From IMSL the subroutines (i) MDFI, and (ii) MDGAM were used.

The values of (5.1) for $\alpha = 0,05$ are given, up to $O(m^{-2})$, for $\lambda = 2, 6, 10, 16, 28, 40$; $m = 50, 100, 1000, \infty$ and $p = 2$, $n = 10$ in table 1. For $p = 2$, $n = 4$ table 2 gives values of (5.1) for $\lambda = 2, 10$, and $m = 50, 100$. Also given are the exact values in this case computed by J. Roy (1965) and the values obtained from the χ^2 -approximation of de Waal (1968). It is evident from these tabulations that the G -values obtained by using equation (5.1) are an improvement over those obtained by using de Waal's asymptotic formula.

TABLE 1

A COMPARISON OF EXACT VALUES, CHI-SQUARE APPROXIMATION, AND
G-APPROXIMATION OF THE POWER OF $|L|$ FOR $p=2$ AND $n=10$ AT 5% LEVEL

λ	m = 50			m = 100			m=1000	m = INF	
	Exact	χ^2	G	Exact	χ^2	G	G	Exact	χ^2
2	0,0874	0,0841	0,0856	0,0913	0,0904	0,0920	0,0974	0,0961	0,0956
6	0,1910	0,1808	0,1833	0,2092	0,2063	0,2090	0,2324	0,2319	0,2293
10	0,3208	0,3053	0,3087	0,3574	0,3532	0,3567	0,4008	0,4019	0,3970
16	0,5257	0,5158	0,5198	0,5835	0,5813	0,5850	0,6451	0,6483	0,6451
28	0,8283	0,8472	0,8495	0,8797	0,8853	0,8871	0,9214	0,9240	0,9200
40	0,9545	0,9712	0,9718	0,9765	0,9805	0,9809	0,9891	0,9898	0,9889

Exact = Roy's power computation,

χ^2 = de Waal's approximation,

G = power computation based on equation (5.1) of this paper.

TABLE 2

A COMPARISON OF EXACT VALUES, CHI-SQUARE APPROXIMATION,
AND G-APPROXIMATION OF THE POWER FUNCTION FOR
 $p = 2$ AND $n = 4$ AT 5% LEVEL

m = 50			
λ	Exact	χ^2	G
2	0,1209	0,1696	0,1209
10	0,5225	0,5201	0,5227

m = 100			
2	0,1257	0,1254	0,1267
10	0,5545	0,5539	0,5563

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